

Cyber Security Lab Report

Experiment 1: OSINT-Based Email Enumeration Using theHarvester

Aim: Perform an OSINT assessment on tesla.com using theHarvester. **Commands:**
theHarvester -d tesla.com -b all
theHarvester -d tesla.com -b bing **Collect:** Public email addresses, subdomains, hosts, IP addresses. **Email Categories:** Support, Sales, Human Resources, Administration, Technical Teams. **Risk Analysis:** Reconnaissance reveals company structure and internet-facing assets. Spear-phishing uses legitimate-looking addresses. Credential harvesting through fake login portals. Impersonation of HR, IT Support, or executives. **Countermeasures:** Reduce public email exposure. Implement SPF, DKIM and DMARC. Enable Multi-Factor Authentication. Conduct regular phishing awareness training. Deploy secure email gateways and monitor exposed assets. **Note:** Insert your own screenshots and actual theHarvester results before submission.

Experiment 2: Phishing Awareness Simulation Using GoPhish

Aim: Conduct a phishing awareness simulation using GoPhish. **Steps:** Start GoPhish (sudo ./gophish). Create sending profile. Create phishing email template. Create landing page. Add dummy employee accounts. Launch campaign and monitor results. **Sample Metrics:** Delivery Rate: 100% Email Open Rate: 80% Click Rate: 35% **Recommendations:** Run periodic phishing simulations. Verify sender addresses. Encourage reporting suspicious emails. Enable MFA. Use secure email gateways. **Note:** Replace sample statistics with your campaign results and screenshots.

Experiment 3: Nmap Aggressive Scan

Aim: Scan scanme.nmap.org using Nmap Aggressive Scan. **Command:**
nmap -A scanme.nmap.org **Collect:** Operating system, service versions, open ports. **Attack Methodology:** Attackers can use service versions and operating system details to impersonate IT support and trick employees into revealing credentials. **Defensive Recommendations:** Close unnecessary ports. Patch operating systems and services. Restrict version disclosure. Use firewalls and IDS/IPS. Train employees against social engineering. **Note:** Insert your actual Nmap output and screenshots.